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Concise Report

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Construction of 1,3-Diboryl Allenes from *gem*-Iododiboron Compounds via Intermolecular Boryl Group Transfer†

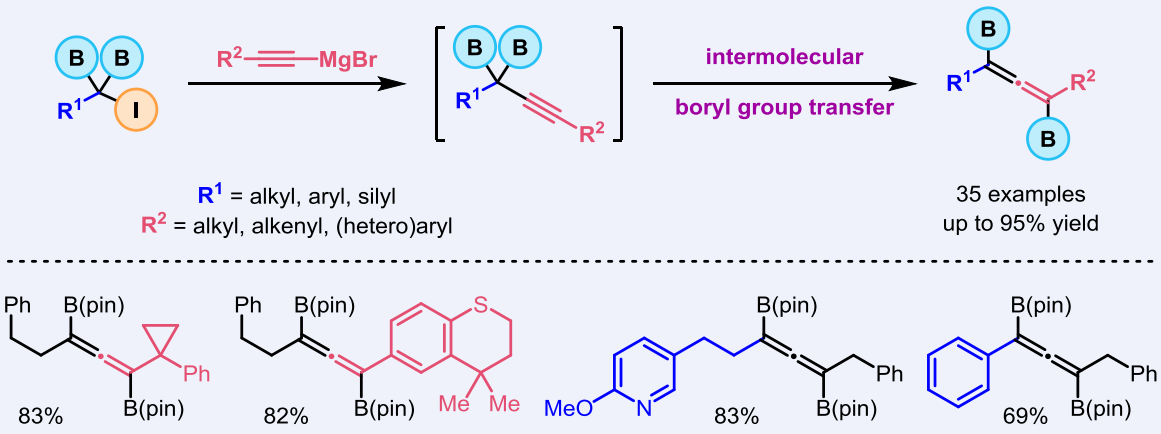
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Keywords

Allenes | Organoboron | Iododiboron | Grignard reagents | Isomerization | Migration | Organohalides | Isotopes | Nucleophilic substitution | Synthetic methods

Comprehensive Summary



Allenylboronates are valuable synthetic building blocks and have garnered significant interest. However, their 1,3-diboyl substituted analogs have not been reported. Herein, we disclosed a novel method for creating 1,3-diboyl allenes from *gem*-iododiboron compounds and alkynyl Grignard reagents. The starting materials are initially converted into the corresponding propargylic *gem*-bis(boronates), which then undergo an intermolecular boryl group transfer to afford 1,3-diboyl allenes, as confirmed by mechanistic studies. This method features good tolerance to steric hindrance and is compatible with a wide variety of functional groups.

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† Dedicated to the Special Issue of Boron Chemistry.

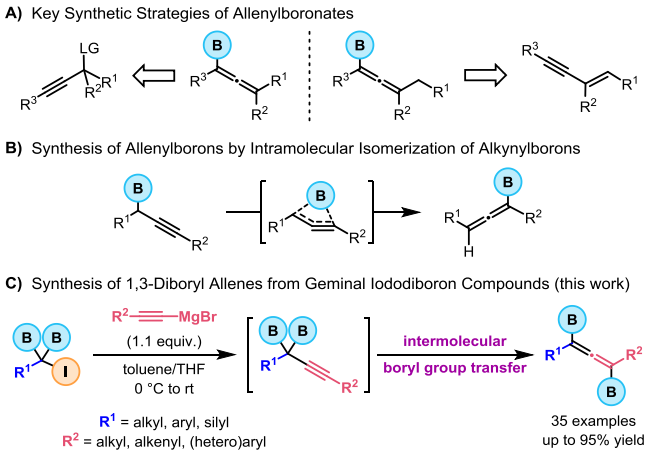
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Background and Originality Content

Allenes are a unique class of unsaturated compounds with a cumulative double-bond skeleton, having widespread applications in natural product synthesis, pharmaceutical chemistry, and materials science.[1] Allenylboron compounds,[2] in particular, have proven themselves to be valuable building blocks in propargylation, allenylation,[3-4] and many other useful transformations.[5] Moreover, the versatile C—B bond can be readily converted into various functional groups.[6] Thus, chemists have persisted in seeking methods for the facile synthesis of allenylboronates from readily available starting materials. Currently, the two major approaches are: 1) boryl substitution of propargylic compounds bearing a leaving group[7] and 2) hydroboration of enynes with HB(pin) or HB(cat)[8] (Scheme 1A). Recently, more strategies, such as olefin metathesis[9] and hydrazination/fragmentation of alkynyl boronates,[10] have been reported. Intriguingly, allenylboronates can also be accessed from the isomerization of propargylic boron compounds,[11-12] featuring an intramolecular boryl group migration as the key step (Scheme 1B). Despite the extensive study of the synthesis and application of allenylboronates, research on diboryl-substituted allenes remains elusive. Given the unique reactivities of the analogous diboryl alkene species,[13] the synthetic protocols for diboryl allenes are highly desired.

Halogenated geminal bis(boronates) have recently emerged as versatile synthons and attracted significant attention.[14] Independent research by Cho,[15] Liu,[16] and Hong[17] groups has led to the development of substitution reactions of halogenated *gem*-diboron compounds with various nucleophiles. While most sp^3- and sp^2-hybridized Grignard reagents were compatible and yielded formal substitution products,[17] we discovered that the sp-hybridized Grignard reagents, *i.e.*, alkynyl magnesium halides, produced 1,3-diboryl allenes predominantly (Scheme 1C). In this paper, we present the development of this method and the mechanistic studies behind this unprecedented transformation.

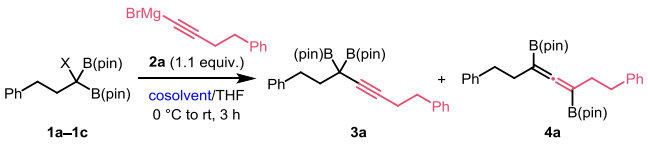
Scheme 1 Synthesis of allenylboronates and 1,3-diboryl allenes



Results and Discussion

Our investigation commenced with the addition of alkynyl Grignard reagent **2a** to an α-halogenated *gem*-diboron compound **1** (Table 1). According to our previous studies,[17] the halogen substituent and the cosolvent were found crucial to the reaction. After initial tests, excellent yield was observed when the *gem*-diboryl iodide **1c** was employed as the substrate (entries 1—3). Due to the lower reactivity of the sp-hybridized carbanion, magnesium-iodine exchange, which was previously observed with aryl and alkyl Grignard reagents, did not occur. To our delight, the propargyl *gem*-diboron product **3a** was not generated, which is otherwise challenging to separate. The addition of toluene as the cosolvent was not necessary compared to pure THF or diethyl ether/THF system (entries 4 and 5). However, with toluene, the reaction was homogeneous, which was preferred for further study. 1,4-Dioxane, DME, and cyclohexane, however, hampered the reaction efficiency and provided **3a** as the major product (entries 6—8). Lastly, alkynyl lithium and zinc reagents were investigated, and the desired product was also observed, albeit in lower yield (entries 9 and 10).

Table 1 Optimization of reaction conditions[a]



Entry	X	Cosolvent	Yield of **3a**/%	Yield of **4a**/%
1	Cl	toluene	0	85
2	Br	toluene	0	87
3	I	toluene	0	92 (86)[b]
4	I	—	0	91
5	I	Et_2O	0	92
6	I	1,4-dioxane	80	10
7	I	DME	47	29
8	I	*c*-hexane	44	0
9[c]	I	toluene	0	76
10[d]	I	toluene	0	80

[a] Conditions: halogenated *gem*-diboron compound **1** (0.1 mmol), cosolvent (1 mL), Grignard reagent (0.11 mmol, 0.7—0.9 mol/L in THF), 0 °C to rt, 3 h. The crude yield was determined by ^{1}H NMR using CH_2Br_2 as the internal standard. [b] Isolated yields in parentheses. [c] Alkynyl lithium reagent was used instead of Grignard reagent. [d] Zinc reagent was used instead.

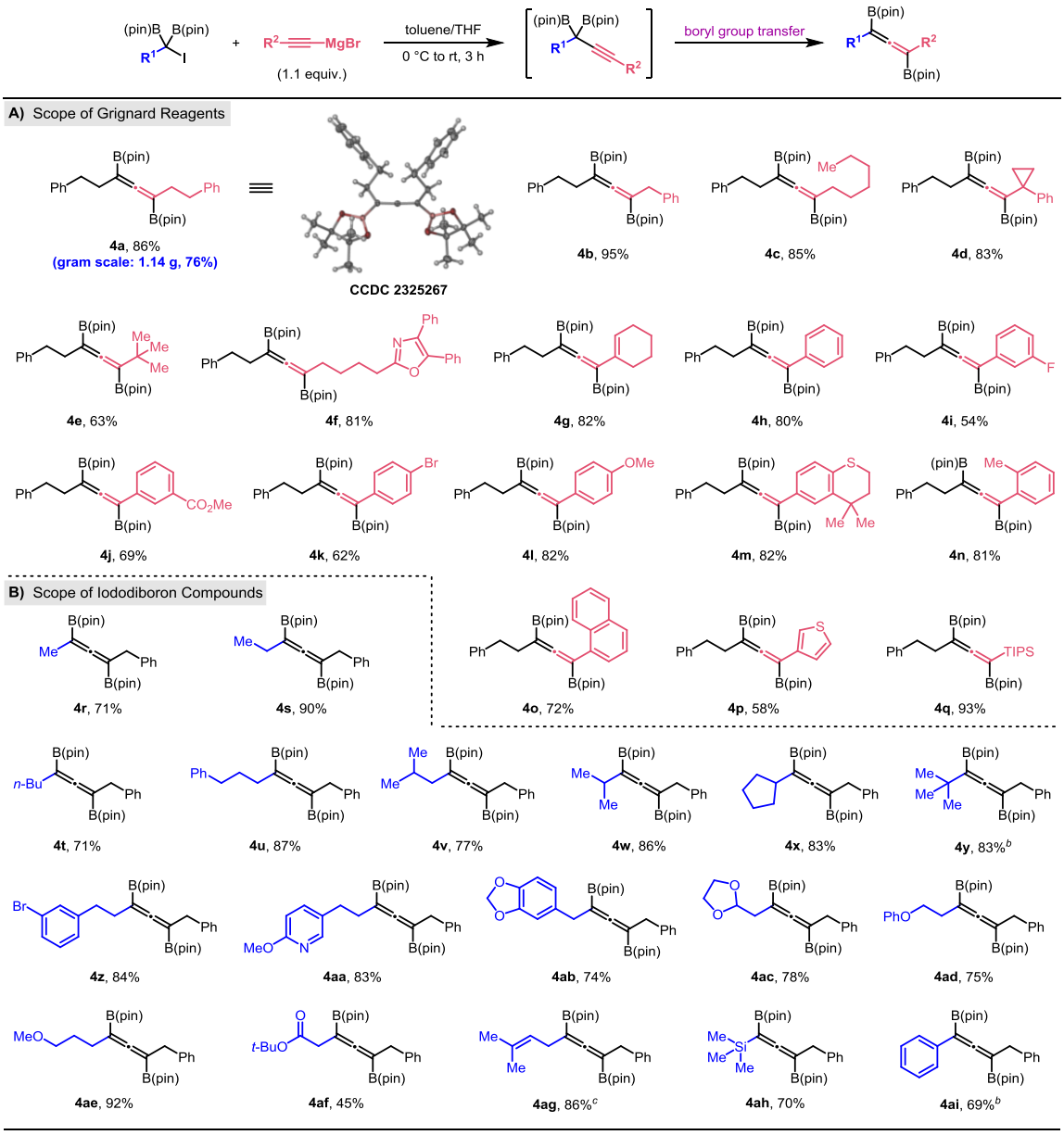
With the optimal condition in hand, the scope of Grignard reagents was explored first using **1c** as the standard substrate (Table 2A). The diboryl allene skeleton of **4a** was confirmed by single crystal X-ray diffraction. A gram-scale experiment was also performed under the standard conditions. The yield slightly diminished due to the prolonged purification on silica gel. This protocol was not sensitive to the steric hindrance on the far side of the alkynyl reagents. Products with both linear chains and hindered quaternary carbon centers were obtained in good yields (**4b—4e**). Grignard reagents bearing alkenyl (**4g**) and aryl groups (**4h—4p**) also performed well. A variety of electron-donating and withdrawing substituents at the *ortho*-, *meta*- and *para*-positions were compatible with little impact on the overall efficiency. Notably, this method was also competent to naphthyl group (**4o**) and aromatic heterocycles (**4f** and **4p**). Additionally, a corresponding product with geminal silyl and boryl substitution (**4q**) was obtained in excellent yield, providing potential for chemoselective transformation.[7a,18]

Next, we examined the scope of the iododiboron compounds using a 3-phenyl-1-propyne-derived Grignard reagent (Table 2B). A wide range of linear and branched alkyl substituents with different sizes and functional groups were surveyed. The products were obtained in high yields in most cases (**4r—4ag**). Notably, the sterically hindered *tert*-butyl group was well tolerated (**4y**). Despite the use of *gem*-diboryl bromide due to the preparation method of the starting material,[19] high efficiency was maintained. Functional groups such as bromo and pyridyl groups, acetal, ether, and allyl groups were all compatible. However, the product possessing an ester group was collected in only 45% yield (**4af**). This inferior result was presumably attributed to the coordination between the carbonyl oxygen and the boron atom, which might

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Table 2 Substrate Scope[a]



[a] Standard condition: iododiboron (0.1 mmol), toluene (1 mL), Grignard reagent (0.11 mmol, 0.5—1.0 mol/L in THF), 0 °C to rt, 3 h. Isolated yields. [b] Bromodiboron was used instead of iododiboron. [c] Chlorodiboron was used instead of iododiboron.

facilitate the enolate formation and interfere with the desired reaction. Lastly, the suitability of silyl (**4ah**) and aryl group (**4ai**) for this protocol was also confirmed.

Our efforts were then devoted to elucidating the reaction mechanism. Initially, substrate **5**, in which one of the B(pin) groups was replaced with a methyl group, was subjected to the standard condition (Scheme 2A). The reaction efficiency decreased significantly, and more importantly, propargyl boronate **6** emerged as the primary product. This outcome aligned with the results in Table 1 (entries 6—8), suggesting that the reaction may involve a formal substitution followed by an isomerization step, both potentially promoted by the geminal diboryl groups. To further validate this hypothesis, we carried out the reaction using iododiboron **1c** at –30 °C, and isolated the corresponding propargyl *gem*-diboron compound **8** as the sole product in 52% yield (Scheme 2B). Subsequently, with the addition of a catalytic amount of Grignard reagent, compound **8** underwent complete isomerization to 1,3-diboryl allene **4b** in 80% NMR yield (Scheme 2C).

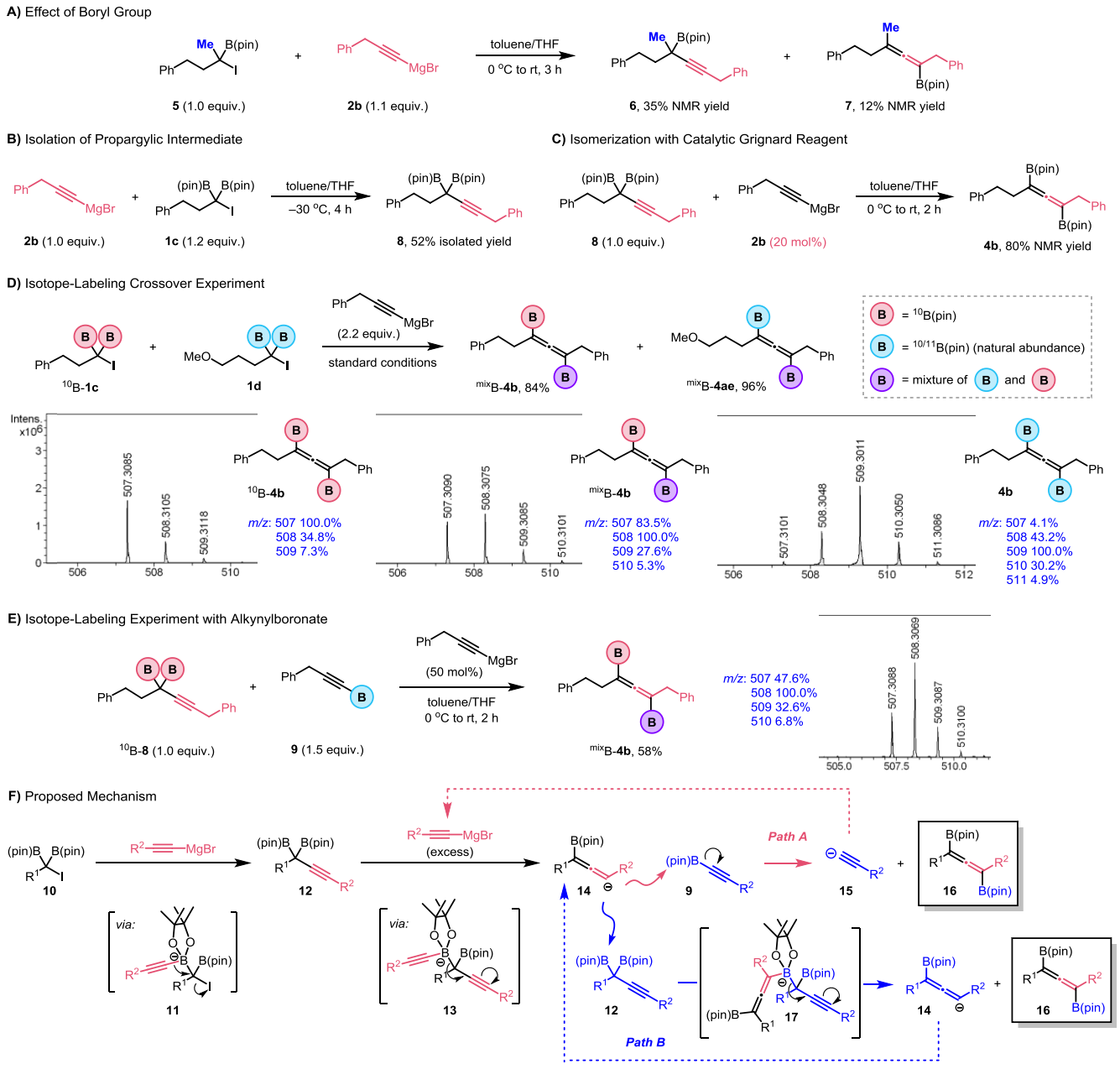
In previous studies by Soderquist and Zhu (Scheme 1B), intramolecular boryl migration was proposed to explain the isomerization.[11] To investigate whether our reaction follows a similar mechanism, we conducted an isotope-labeling crossover experiment (Scheme 2D). We initially combined ^{10}B-labeled **1c**[20] with unlabeled **1d** in a 1 : 1 ratio and then exposed the mixture to the standard condition. The resulting products, **4b** and **4ae**, were isolated and analyzed using high-resolution mass spectroscopy (HRMS). Upon comparing the spectra of the collected product **4b** to that of authentic ^{10}B-**4b** and unlabeled **4b** with natural abundance, it became evident that there was a scrambling of boryl groups between ^{10}B-**1c** and **1d**. The same phenomenon was also observed for **4ae** (for HRMS spectra, see Supporting Information). This outcome strongly indicated that an *intermolecular* pathway is the predominant, if not the sole, route during the isomerization.

To provide additional mechanistic insights, we performed another isotope-labeling experiment. Specifically, we conducted the isomerization of ^{10}B-**8** in the presence of a stoichiometric amount of unlabeled alkynyl boronate **9** (Scheme 2E). The HRMS spectrum of product ^{mix}B-**4b** further confirmed the involvement of an alkynyl boronate species in the major pathway.

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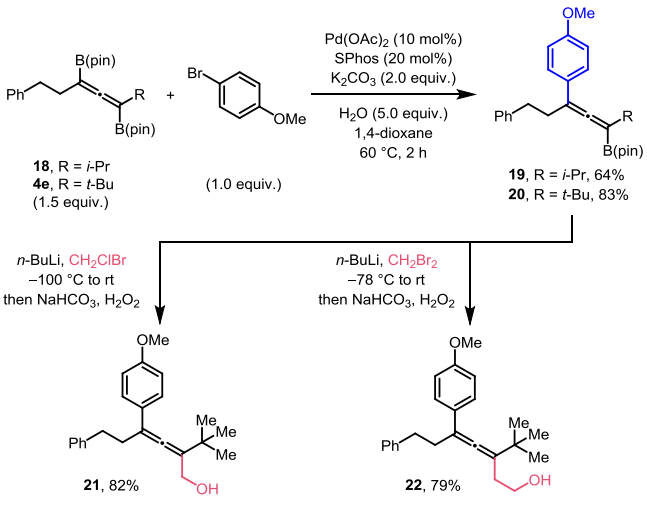
Scheme 2 Mechanistic studies



In light of the mechanistic experiments and our previous studies,[17] a plausible mechanism is proposed in Scheme 2F. Substrate **10** was initially converted into *gem*-diboryl alkyne **12** through the formation of ate-complex **11**, followed by a 1,2-metallate shift. The coordination of compound **12** with excess Grignard reagent generates ate-complex **13**, and the cleavage of the C—B bond leads to the formation of a more stable sp^2-hybridized carbanion **14**. This carbanion can serve as a Lewis base to abstract a boryl group from alkynyl boronate **9** to afford the desired product **16**. The resulting alkynyl carbanion **15** can then activate another molecule of **12** in a catalytic manner (Path A with red arrows). The alternative pathway involves a boryl transfer from another molecule of **12** through the intermediacy of ate-complex **17**, yielding compounds **16** and **14**, which can then sustain a chain process (Path B with blue arrows). Based on the current information, the occurrence of Path A can be confirmed. However, both pathways appear to contribute to the reaction, and it remains unclear which of them is predominant.

Lastly, the synthetic application of the diboryl allene products was briefly demonstrated in Scheme 3. The regioselective Suzuki-

Scheme 3 Synthetic Application



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Miyaura cross-coupling reactions were conducted successfully using sterically biased substrates such as **18** and **4e**. Product **20** could undergo either mono- or di-homologation under precisely tuned conditions,[21] followed by oxidation to provide allenyl or homoallenyl alcohols, respectively.

Conclusions

In summary, we have disclosed the first example of accessing 1,3-diboryl allenes from *gem*-iododiboron compounds in the presence of alkynyl Grignard reagents. The reaction is highly efficient and can accommodate steric hindrance and various functional groups. Mechanistic studies have suggested that the cascade process consists of a formal substitution and a subsequent isomerization. The isotope-labeling crossover experiment indicated that the isomerization proceeds through an unconventional intermolecular boryl group transfer. Utilization of the products was also briefly demonstrated.

Experimental

In an argon-filled glove box, to a 15 mL vial equipped with a magnetic stir bar was added the iododiboron compound (0.1 mmol, 1.0 equiv.). The vial was sealed with a polypropylene open-top cap with PTFE/silicone septum, then removed from the glove box. Toluene (1.0 mL) was added and the vial was cooled in an ice-cold water bath. Grignard reagent (0.11 mmol, 1.1 equiv.) was slowly added. The mixture was then warmed to room temperature and stirred for 3 h. Upon completion, the reaction was quenched with sat. NH_4Cl solution (2 mL) and extracted with Et_2O (3 x 2 mL). The combined organic layers were dried over Na_2SO_4, filtered, and concentrated *in vacuo*, and the crude reaction mixture was purified by column chromatography on *pre-dried* silica gel.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202400840.

Acknowledgement

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